

# Dr Jinyong (Edward) Sung

Mobile: (424)-787-066 | Email: edward.sung@innofusality.com | Profile: <https://innofusality.com/>

## Profile

---

- ✓ ACU APTT Scholar pursuing a Master of Teaching (Secondary) in **Economics & Mathematics**; Former UTS Research Excellence Scholar / Consultant with expertise in **empirical economics, computational modelling, and policy analysis** at UTS ISF
- ✓ +10 years of work experience in **corporate/project finance and management** consulting at **Global energy/financial companies** and the **U.S. Top-tier/Ivy League education**; and Bilingual in English and Korean
- ✓ Core competencies: Econometric & Computational Modelling | Machine & Deep Learning | Policy & Financial Analysis

## Education

---

**Australian Catholic University, Faculty of Education and Arts** Sydney, NSW, Australia  
Master of Teaching (Secondary) – Economics & Mathematics, All Points to Teaching (APTT) Scholar Mar 2026 – Present  
*Pre-service teacher through the Australian Government High Achieving Teachers Program (employment-based pathway)*

**University of Technology Sydney, Institute for Sustainable Futures** Sydney, NSW, Australia  
Ph.D. in Sustainable Futures – Energy, specialising in Energy Economics and Policy & Modelling Feb 2022 – Feb 2026  
*Focus on integrating machine & deep learning with econometric & economic modelling for energy economics and policy*

- Supervisory panel: 1. [Principal supervisor] Prof. **Xunpeng (Roc) Shi** - PhD in Economics, UTS ACRI  
2. [Co-supervisor] Prof. **Sven Teske** - PhD in Engineering & Economics, UTS ISF  
3. [Research advisor] Dr. **Mengheng Li** - PhD in Econometrics, Senior Lecturer, UTS Business School
- Research field subject result: Econometrics (Grade: High Distinction)

**Johns Hopkins University** Baltimore, MD / Washington, DC., USA  
MBA/MA Dual Degree, specialising in Energy & Finance Aug 2010 - May 2013

- M.A. in International Economics & IR, (Energy Focus) – School of Advanced International Studies (SAIS)
- MBA (Finance; Energy & Environment) – Carey Business School

**Brown University, School of Engineering** Providence, RI, USA  
M.S. in Innovation Management and Entrepreneurship Engineering May 2008

**Purdue University, College of Engineering** West Lafayette, IN, USA  
B.S. in Industrial Engineering (Major) & Mathematics (Minor) May 2007  
*Completed 4-year program in 2 years through transfer credits from a top-tier Korean engineering university and Purdue Wave Exams.*

## Fields of interest

---

Energy Economics & Policy, Computational & Applied Economics, Machine Learning & Deep Learning, Sustainability

## Research

---

### Publications

- Sung, J., Shi, X., Teske, S., & Li, M. (2024). Chinese natural gas phase-out pathways: A novel hybrid scenario-specific projection approach to achieve Net Zero. *Accepted, Energy, 2025*
- Sung, J., Shi, X., Teske, S., & Li, M. (2025). Identifying key natural gas drivers towards China's carbon neutrality through a novel temporal shrinkage framework. *Accepted, Resources Policy, 2026*
- Sung, J. (2025). Opportunities for Australia amid China's carbon-neutral transition: Input–Output approach with analysis of Australian renewable potential. *Accepted, Sustainable Development, 2026*

### Working papers

- Sung, J., Shi, X., Teske, S., & Li, M. (2025). Carbon-neutral planning framework for China's near-term natural gas demand: A neural-enhanced hybrid statistical–deep learning forecasting approach. *Submitted (Revise & Resubmit)*

### Non-academic work

- Shi, X. & Sung, J. (2023). Australia's natural gas security policy spills over the region. *East Asia Forum post*

## Awards

---

- ACU All Points to Teaching (**APTT**) **Scholarship** (awarded to high-achieving pre-service secondary teachers), 2026–2027
- Australian Government Research Training Program Stipend (**UTS Research Excellence Scholarship** for highly ranked, newly-

commencing doctoral students) / Fee Offset Scholarship, UTS, 2022-2025

- **Vice Chancellor's Conference Fund 2024** of the 41<sup>st</sup> USAEE/IAEE Conference, UTS, 2024
- **Certificate of Appreciation** in recognition of high-performing work at SK advanced (SK Group's Chemical arm), 2017
- **SAIS Summer Internship Fund**, Johns Hopkins School of Advanced International Studies, 2012
- **Research Assistantship**, Johns Hopkins Carey Business School, 2010F, 2011S

## **Professional experience**

---

St Patrick's Marist College Dundas

Sydney, NSW, Australia

### **Student Teacher (Teaching Practicum)**

May 2026

- Planned and Delivered lessons in Year 12 Economics, Year 11 Business Studies, and Year 9 Mathematics

UTS Institute for Sustainable Futures

Sydney, NSW, Australia

### **Research Consultant**

May 2025 – June 2026

- Undertaking economic research of an organic waste-based bioenergy project, developing its financial models, and analysing its financial viability through analysis of CAPEX, OPEX, ROI, revenue forecasts, and sensitivity scenarios.

### **PhD researcher**

Feb 2022 – Feb 2026

- Conducted interdisciplinary research on the energy transition in the Australia–China natural gas sector, leading to multiple academic publications and working papers using advanced econometric and machine/deep learning techniques.

EMS Trust (specializing in investing in education biz & commercial/residential properties)

Sydney, NSW, Australia

### **Investment Director & Co-founder**

Dec 2018 – Dec 2021

- Developed investment opportunities across commercial and residential property, as well as the education sector; operated the Trust and devised investment strategies in e-learning and online business.
- Built management system for education business; managed finance and human resources; created comprehensive in-house system—including invoicing—using Excel and financial modelling.

Mazars Global Infrastructure Finance Pty Ltd

Sydney, NSW, Australia

### **Project Manager, Model Build Team**

Oct 2017 – Nov 2018

- Built and audited financial models for energy investments; delivered financial modelling training to over 100 professionals across Asia and Australia, enhancing analytical capabilities within investment teams.
- Modelled renewable energy projects (solar and wind) in Australia and internationally, oil and gas (O&G) projects in the global upstream/midstream sector, a public-private partnership (PPP) project in Gold Coast, Australia, a smelting project in Indonesia, and a corporate model for a mobile services company in Australia.
- Acted as the communication channel with Mazars Seoul office for potential energy and infrastructure projects in South Korea.

SK Advanced (Chemical arm of SK Group; 3-party JV: APC(KSA Chem)-PIC(Kuwaiti Chem)-SK Gas)

Seoul, S. Korea

### **Strategy & Planning Manager, Management Planning Team**

Mar 2015 – Oct 2017

- Demonstrated people management by aligning diverse stakeholder interests—including board members and shareholders—to support strategic decision-making; conducted quantitative analysis for infrastructure asset acquisitions.
- Devised a joint venture (JV) implementation plan—including valuation—to establish a three-party JV with two Middle Eastern chemical companies, APC (Saudi Arabia) and PIC (Kuwait), resulting in the successful formation of the JV.
- Applied people management skills in coordinating with JV board members from Korea, Saudi Arabia, and Kuwait, and crafted key agreements and business systems of the JV.
- Facilitated in-house corporate business training for junior staff and new joiners.

SK Gas (Energy arm of SK Group, 3<sup>rd</sup> Korean conglomerate/top 100 in Fortune Global 500)

Singapore/Seoul, S. Korea

### **Strategy & Planning Manager (Employed by SK Global MBA recruitment), Corporate Strategy team**

Jun 2013 – Mar 2015

- Revamped financial and economic analytics for \$1B petrochemical (PDH) business; established merchant LPG trading model in collaboration with trading teams in Houston and Singapore.
- Performed due diligence for \$4B, 2GW coal-fired power plant JV; initiated valuation analyses for 500 MW LPG power plant in Honduras and 1 MW solar and fuel cell power plants in Korea and the U.S.
- Devised long- and mid-term LPG logistics optimisation plan; evaluated impact of declining oil prices on overall business strategy.

Center for Climate and Energy Solutions (Top-tier U.S. think tank on Global Climate Change)

Arlington, VA, USA

### **Policy Analyst (MBA/MA Summer Intern), International Policy Research Team**

Jun 2012 – Aug 2012

- Conducted research on market-based carbon policy instruments in Brazil, China, South Africa, and South Korea, informing global carbon market insights.

Johns Hopkins Carey Business School

Baltimore, MD, USA

### **Research Assistant under Prof. Mitsukuni Nishida (awarded a Research Assistantship)**

Sep 2010 – May 2011

- Conducted modelling of economic development and utility infrastructure growth correlations across emerging Asian countries.

Choong Ang Biotech Co., Ltd

Seoul, S. Korea

### **Equity Financing Project Manager, Strategic Planning Team**

Dec 2009 – May 2010

- Led a \$13M paid-in capital increase aligned with the company's equity financing goals.
- Developed equity financing strategies and supervised investor presentations for shareholders and key prospective investors, resulting in the attraction of a \$6M investment fund.

Hanwha Investment & Securities (Financial arm of Hanwha Group, 10<sup>th</sup> Korean conglomerate)

Seoul, S. Korea

### **Associate, Investment Banking II Team (Employed by Hanwha Global recruitment)**

Jun 2008 – Nov 2009

- Conducted due diligence, developed capital financing strategies, and delivered presentations to CEOs, executives, and shareholders, which resulted in 10 successful capital financing agreements.
- Authored 20 capital financing prospectuses and managed post-issuance processes for major equity and bond transactions, including Hanwha's \$290M capital increase, Hanil's \$50M raise, and multiple bond and CB issuances.
- Demonstrated expertise as a financial advisor by advising on an \$11M M&A project (backdoor listing).

NAVER Corporation (Korean internet conglomerate)

Seoul, S. Korea

### **Summer Intern (Employed by the U.S. Internship Recruitment), Market Strategy Team**

Jun 2007 – Aug 2009

- Analysed competitor marketing strategies and assessed societal implications of digital advertising.

## **Professional training and coaching experience**

---

### **Student Teacher (Teaching Practicum), St Patrick's Marist College Dundas**

May 2026

- Designed and delivered curriculum-aligned lessons in Economics (Year 12), Business Studies (Year 11), and Mathematics (Year 9)

### **Financial modelling trainer, Mazars Global Infrastructure Finance Pty Ltd**

Oct 2017 – Nov 2018

- Delivered training courses in Best Practice Project Finance Modelling and Excel Diploma

### **New employee trainer, SK Advanced**

Mar 2015 – Mar 2017

- Understanding of the Propane De-Hydrogenation (Petrochemical) business

## **Presentations**

---

The 3rd ISETS Annual Conference in Tokyo, Japan

2025

The 41st USAEE/IAEE Annual Conference in Baton Rouge, LA, USA; The 2nd ISETS Annual Conference in Nanjing, China

2024

The 34th CESA Annual Conference in Adelaide, Australia; Seminar (invited guest speaker) in Beijing Institute of Technology China; Seminar (invited guest speaker) in Yonsei University, South Korea; Knowledge Sharing session in UTS ISF

Knowledge Sharing session in ACRI

2023

Knowledge Sharing session in UTS Australia-China Relations Institute (ARCI)

2022

## **Postgraduate (PhD) training**

---

Advanced-level training in Dynamic CGE modeling using GTAP/GEMPACK, *M&S Research Hub, Germany*

Jun 2023

Introductory to Intermediate-level training in CGE modeling using GTAP/GEMPACK, *M&S Research Hub, Germany*

May 2023

Time Series Analysis, *UTS Economics Discipline Group*

Aug 2022 – Dec 2022

Econometrics, *UTS Economics Discipline Group*

Aug 2022 – Dec 2022

## **Certifications and Professional Recognition**

---

Certificate of Achievement: Introduction to Programming with Python (CS101), KAIST

Mar 2024

Certificate of Achievement: Practical Python for AI Coding, KAIST

Jun 2023

Management Consulting Professional (Validated by VETASSESS – Australian Government, ANZSCO 224711)

Jun 2017

## **Additional information**

---

Skills: **Python - Deep/Machine Learning (Master), MS Excel/VBA (Master)**, Matlab, MS PowerPoint, GAMS, Power BI, HTML

Citizenship & Residency status: South Korean citizen and Australian permanent resident

References:

### **Professor Xunpeng (Roc) Shi**

Australia-China Relations Institute, UTS

UTS Building 1, 18/15 Broadway,

Ultimo, NSW 2007, Australia

Email: Xunpeng.Shi@uts.edu.au

Phone: +61 2 9514 8953

### **Professor Sven Teske**

Institute for Sustainable Futures, UTS

UTS Building 10, 235 Jones St, Ultimo,

NSW 2007, Australia

Email: Sven.Teske@uts.edu.au

Phone: +61 2 9514 4950

### **Dr Mengheng Li, Senior Lecturer**

Economics Discipline Group, UTS

UTS Building 8, 14 - 28 Ultimo Rd,

Ultimo, NSW 2007, Australia

Email: Mengheng.Li@uts.edu.au

Phone: +61 2 9514 3074